

SOCIAL NETWORKS

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Week 4 — Question Paper (Comprehension-Based MCQ / MSQ)

Comprehension 1 — Birds of a Feather: Homophily, Selection and Influence

Case Study

A campus wellness team studies how a student community forms friendships and how habits spread through it. Two forces interest them. Sometimes people befriend others who are already similar to them; sometimes people first become friends and then grow similar by picking up each other's habits. The team also knows that friendships don't form only through mutual friends — shared activities and group memberships pull people together too.

They recall a striking result from the course: traits such as obesity, and even happiness, can behave like contagions, spreading along the friendship network rather than staying with one person. To quantify how alike two people are, they use a simple overlap measure: the number of interests the two share divided by the total number of distinct interests between them. They also want to test, for a whole network, whether similar people befriend each other more often than chance would predict.

For their worked examples they keep two cases in mind. First, two students Anita and Bina: Anita's interests are {chess, hiking, jazz} and Bina's are {hiking, jazz, baking, cycling}. Second, a club of 20 students made of 10 first-years and 10 final-years, with 30 friendships in total, of which 9 join a first-year to a final-year.

Questions 1–10

Q1. [MCQ] The proverb “birds of a feather flock together” names which phenomenon, as defined in the course?

- A. Homophily — the tendency of similar people to form friendships with one another.
- B. Betweenness — the tendency of certain edges to lie on many shortest paths.
- C. Triadic closure — the tendency of a graph to fill in until it is fully connected.
- D. Embeddedness — the count of common friends shared along a single edge.

Answer: A

Reference: Lecture 44 – Homophily: Definition and measurement (≈01:31)

Solution: Homophily is the principle that people who are alike tend to befriend one another — exactly the “birds of a feather” proverb. Betweenness, triadic closure, and embeddedness are different structural notions from earlier in the course.

Q2. [Match the following] Each scenario in Column A illustrates one mechanism in Column B. Match them.

Column A (scenario)	Column B (mechanism)
1. Two students who both already speak French start talking and become friends	a. Membership closure

2. After becoming friends with a smoker, a student takes up smoking	b. Selection
3. Two strangers who attend the same yoga class become friends	c. Foci closure
4. A student brings an existing friend along to her music club	d. Social influence

- A. 1-d, 2-b, 3-a, 4-c
- B. 1-b, 2-d, 3-c, 4-a
- C. 1-b, 2-d, 3-a, 4-c
- D. 1-a, 2-c, 3-d, 4-b

Answer: B (1-b, 2-d, 3-c, 4-a)

Reference: Lecture 42 – Selection and Social Influence; Lecture 45 – Foci and Membership Closure

Solution: Befriending an already-similar person is selection (1-b). Changing a habit because of a friend is social influence (2-d). Meeting through a shared activity/focus is foci closure (3-c). Bringing a friend into a focus you belong to is membership closure (4-a).

Q3. [MCQ] Using the overlap (similarity) measure, how similar are Anita {chess, hiking, jazz} and Bina {hiking, jazz, baking, cycling}?

- A. $2/7 \approx 0.29$
- B. $2/3 \approx 0.67$
- C. $3/4 = 0.75$
- D. $2/5 = 0.40$

Answer: D

Reference: Lecture 43 – Interplay between Selection and Social Influence (≈03:22–04:35)

Solution: Similarity = (shared interests) ÷ (all distinct interests). Shared = {hiking, jazz} = 2; the union {chess, hiking, jazz, baking, cycling} = 5. So the measure is $2/5 = 0.40$.

Q4. [MCQ] For the club of 20 (10 first-years, 10 final-years) with 30 friendships, 9 of which are cross-year, homophily is computed as $1 - (\text{actual cross-year edges}) \div (\text{expected cross-year edges})$. With a 50–50 split the expected cross-year fraction is one-half. What is the homophily value?

- A. $1 - 9/15 = 0.40$
- B. $1 - 9/30 = 0.70$
- C. $1 - 9/9 = 0.00$
- D. $9/30 = 0.30$

Answer: A

Reference: Lecture 44 – Homophily: Definition and measurement (≈09:25–13:03)

Solution: With a 50–50 split, a random friendship is cross-year half the time, so the expected number of cross-year edges is half of 30 = 15. The measure is $1 - \text{actual/expected} = 1 - 9/15 = 1 - 0.6 = 0.40$. A positive value indicates homophily — fewer cross-group ties than chance.

Q5. [MCQ] How should the sign of the measure $1 - (\text{actual} \div \text{expected})$ cross-group edges be interpreted?

- A. Positive means heterophily; zero means homophily; negative means random mixing.
- B. Positive means homophily; zero means chance-level mixing; negative means heterophily.
- C. The value always stays between 0 and 1, and larger always indicates more random mixing.
- D. Only the magnitude of the value matters; its sign carries no real meaning at all.

Answer: B

Reference: Lecture 44 – Homophily: Definition and measurement (≈13:03–14:31)

Solution: Fewer cross-group edges than expected makes the fraction small and the measure positive — homophily. If cross-group edges match expectation, the measure is 0 — random mixing. If there are more cross-group edges than expected (as in the actor–actress party), the measure goes negative — heterophily.

Q6. [MCQ] In the Wikipedia study of editors' similarity over time around their first interaction, what distinguishes selection from social influence?

- A. Selection is the rise in similarity after they first interact; influence is the rise before it.
- B. Selection and influence are identical and cannot be told apart from the similarity curve.
- C. Selection is the rise before contact; influence the rise after; on Wikipedia selection was steeper.
- D. Neither selection nor influence produces any measurable change in the similarity curve.

Answer: C

Reference: Lecture 43 – Interplay between Selection and Social Influence (≈13:49–15:21)

Solution: Plotting average similarity against time around first contact, similarity climbs before contact — similar people select each other to interact — and climbs further after, as interaction makes them more similar (influence). On Wikipedia the pre-contact (selection) rise was steeper than the post-contact (influence) rise; other datasets, like obesity, can show the reverse.

Q7. [MCQ] The team recalls the contagion findings (Christakis and Fowler). Which statement is correct?

- A. Only obesity is contagious; happiness was shown not to spread through the network at all.
- B. Happiness spreads only to a person's direct friends and reaches no one beyond them.
- C. Contagion effects appeared only in online networks and never in real friendships.
- D. Both obesity and happiness are contagious; happiness was found to spread up to three levels out.

Answer: D

Reference: Lecture 41 – Introduction to Homophily (≈01:40–04:23)

Solution: The longitudinal studies found obesity to be contagious and happiness to spread through the friendship network up to three levels out — your happiness is associated with your friends', their friends', and their friends' friends'. This is precisely what made the result so striking.

Q8. [MCQ] What distinguishes foci closure from membership closure?

- A. Foci closure: a friend joins your club; membership closure: two strangers meet at the same club.
- B. Foci closure: people meet through a shared focus; membership closure: you join a friend's focus.

- C. They are simply two different names for exactly the same single closure mechanism.
- D. Both require a common friend and are really just special cases of triadic closure.

Answer: B

Reference: Lecture 45 – Foci Closure and Membership Closure (≈00:47–02:31)

Solution: Foci closure forms a friendship between two people because they share a focus (a club, a class). Membership closure runs the other way: a person becomes a member of a focus because a friend already belongs. Triadic closure is the third, distinct kind — friendship via a common friend.

Q9. [MSQ] Which statements about selection and social influence are correct? (Select all that apply.)

- I. Selection means choosing friends who are already similar to you.
 - II. Social influence means becoming similar to friends after the friendship forms.
 - III. Two people of the same social status becoming friends is an example of selection.
 - IV. Taking up a friend's habit of eating junk food after befriending them is an example of selection.
- A. I, II, IV
 - B. II, III, IV
 - C. I, III
 - D. I, II, III

Answer: D (statements I, II, III)

Reference: Lecture 42 – Selection and Social Influence

Solution: Selecting already-similar friends (I) and growing similar after befriending (II) are the definitions; same-status people befriending is selection because the similarity precedes the friendship (III). Picking up a friend's junk-food habit happens after the friendship forms, so it is social influence, not selection — making IV false.

Q10. [MSQ] Which statements about homophily and closures are correct? (Select all that apply.)

- I. A homophily value of 0 means cross-group friendships occur at exactly the chance-expected rate.
 - II. Triadic, foci, and membership closure are three distinct ways friendships (or memberships) can form.
 - III. A negative homophily value indicates heterophily — more cross-group ties than chance.
 - IV. Homophily can only ever be measured on networks split exactly 50–50 into two groups.
- A. I, II, III
 - B. I, II, IV
 - C. II, III, IV
 - D. I, IV

Answer: A (statements I, II, III)

Reference: Lecture 44 – Homophily; Lecture 45 – Closures

Solution: A measure of 0 is chance-level mixing (I), the three closures are distinct formation routes (II), and a negative value is heterophily (III). The 50–50 split just makes the expected cross-group fraction a clean one-half; the measure generalises to any split using the appropriate expected fraction, so IV is false.

Comprehension 2 — The Fatman Evolutionary Model

Case Study

To watch homophily, closure and social influence act together, a student builds a simulated city in NetworkX. The city starts as 100 “person” nodes, each given a random BMI, plus five “focus” nodes — a gym, an eat-out place, a movie club, a karate club, and a yoga club. Each node carries two attributes: name (which holds the BMI for a person, or the focus’s label) and type (either 'person' or 'foci'). Initially every person is linked to exactly one randomly chosen focus; foci are never linked directly to one another.

Three rules then drive the evolution. Homophily: two people are more likely to befriend when their BMIs are close, with the connection probability falling as the BMI gap grows. Closure: two nodes are more likely to link the more common neighbours they share, where a shared neighbour can be a person (triadic closure) or a focus (foci/membership closure). Social influence is captured indirectly through the foci — each round, a person attached to the gym loses a little weight while a person attached to the eat-out place gains a little, and membership closure keeps pulling friends into the same foci.

Probabilities are realised with the usual trick: draw a uniform random number in $[0,1]$ and act only if it falls below the target probability. The model runs for several rounds, each round applying homophily, then closure, then the weight change, and the network is saved after each round so its evolution can be analysed later. Key fragments:

```
import networkx as nx, random, math

# assign a random BMI and a type to each person
G.node[v]['name'] = random.randint(15, 40)
G.node[v]['type'] = 'person'

# homophily: closer BMIs -> higher link probability
p = 1.0 / (abs(G.node[u]['name'] - G.node[v]['name']) + 1000)
if random.random() < p:
    G.add_edge(u, v)

# closure: k common neighbours -> probability 1-(1-p)^k
k = len(set(G.neighbors(u)) & set(G.neighbors(v)))
prob = 1 - math.pow(1 - p, k)
```

Questions 1–10

Q1. [MCQ] How is each person node initialised in the model?

- A. name is set to 'person', while the type attribute holds a random BMI value.
- B. name holds a random integer BMI from 15 to 40, and type is set to 'person'.
- C. Both name and type are set to the very same random BMI integer value.
- D. name is a random float in $[0,1]$, and the type attribute is left unused.

Answer: B

Reference: Lecture 47 – Fatman Model: Adding people (≈05:16–08:31)

Solution: Each person gets `name = random.randint(15, 40)` (its BMI) and `type = 'person'`. The attribute `name` deliberately holds the BMI for a person and the label for a focus, so a single attribute is meaningful for both node kinds; `type` distinguishes the two.

Q2. [MCQ] Homophily uses $p = 1 / (| \text{BMI}_u - \text{BMI}_v | + 1000)$. For two people with BMIs 22 and 27, what is p , and why is the +1000 there?

- A. $p = 1/1005 \approx 0.000995$; the +1000 avoids divide-by-zero and scales every probability down.
- B. $p = 1/5 = 0.2$; the +1000 is what makes equal-BMI pairs certain to connect each round.
- C. $p = 5/1000 = 0.005$; the +1000 is included in the formula to raise the probability.
- D. $p = 1000/5 = 200$; the +1000 is itself the connection probability between the pair.

Answer: A

Reference: Lecture 49 – Fatman Model: Implementing Homophily (≈00:53–03:16)

Solution: $|22 - 27| = 5$, so $p = 1/(5 + 1000) = 1/1005 \approx 0.000995$. The +1000 serves two purposes: it prevents a divide-by-zero when two BMIs are equal (gap 0), and it shrinks every probability so the network gains edges gradually rather than all at once.

Q3. [MCQ] The code does `if random.random() < p: G.add_edge(u, v)`. Why does this realise an event of probability p ?

- A. Because `random.random()` always returns exactly the value p on each call.
- B. Because `random.random()` yields integers, which only rarely fall below p .
- C. Because a uniform draw in $[0,1]$ lands below p with probability exactly p .
- D. Because the strict comparison quietly rounds the value of p up to one.

Answer: C

Reference: Lecture 49 – Fatman Model: Implementing Homophily (≈03:16–04:55)

Solution: `random.random()` draws a uniform real number in $[0,1]$. The chance it lands in the sub-interval $[0, p)$ is exactly p , so adding the edge only when the draw is below p makes the edge appear with probability p — the standard way to simulate a probabilistic event.

Q4. [MCQ] Closure uses probability $1 - (1 - p)^k$, where p is one common neighbour's contribution and k is the number of common neighbours. If $p = 0.1$ and two people share $k = 3$ common neighbours, what is the connection probability?

- A. 0.300
- B. 0.729
- C. 0.100
- D. 0.271

Answer: D

Reference: Lecture 50 – Quantifying Triadic Closure; Lecture 51 – Implementing Closures

Solution: $1 - (1 - 0.1)^3 = 1 - 0.9^3 = 1 - 0.729 = 0.271$. More common neighbours raise the probability (each independently 'fails' to connect them with probability $1 - p$, and all k must fail for them to stay unconnected).

Q5. [MCQ] In the closure code, $k = \text{len}(\text{set}(G.\text{neighbors}(u)) \& \text{set}(G.\text{neighbors}(v)))$. What does k count?

- A. The count of neighbours shared by u and v (their neighbour-set intersection).
- B. The combined total of all the neighbours of u and of v taken together.
- C. The total number of edges that are present in the whole graph G .
- D. The shortest-path distance that separates the two nodes u and v .

Answer: A

Reference: Lecture 51 – Fatman Model: Implementing Closures (≈09:57–11:33)

Solution: Converting each neighbour list to a set and taking $\&$ gives the intersection — the nodes adjacent to both u and v . Its length is the count of common neighbours, which feeds the closure formula. A common neighbour may be a person (triadic closure) or a focus (foci/membership closure).

Q6. [MCQ] When applying closure, why must at least one of the two chosen nodes be a person (i.e. both cannot be foci)?

- A. Because foci nodes are never assigned any name attribute in the first place.
- B. Because two person nodes can never share any common neighbour between them.
- C. Because the whole graph would otherwise immediately become completely disconnected.
- D. Because foci are never linked to each other, so two foci can't close into a friendship.

Answer: D

Reference: Lecture 51 – Fatman Model: Implementing Closures (≈03:38–05:03)

Solution: Foci are only ever connected to persons, never to each other, so a friendship edge between two foci is meaningless in the model. Closure is therefore applied only to pairs where at least one node is a person — person–person (triadic, or foci closure via a shared focus) or person–foci (membership closure).

Q7. [MCQ] How is social influence implemented through the foci each round?

- A. Every person's BMI is reset to a brand-new random value at the start of each round.
- B. Every person attached to any focus at all gains one unit of BMI each round.
- C. Gym neighbours lose 1 BMI (floored at 15); eat-out neighbours gain 1 BMI (capped at 40).
- D. Foci nodes change their own BMI based on the members currently attached to them.

Answer: C

Reference: Lecture 52 – Fatman Model: Implementing Social Influence (≈00:20–03:18)

Solution: Each round, neighbours of the gym lose 1 BMI (floored at 15) and neighbours of the eat-out place gain 1 BMI (capped at 40). Combined with membership closure pulling friends into the same foci, this indirectly produces social influence — your friend joins your gym and starts losing weight too.

Q8. [Arrange in order] Arrange the steps of building and running the model in the correct order.

Steps (in scrambled order):

- (i) Add the five focus nodes (gym, eat-out, movie, karate, yoga).
- (ii) Create the graph and add 100 person nodes.

- (iii) Each round, apply homophily, then closure, then change BMIs (social influence).
- (iv) Link each person to exactly one randomly chosen focus.
- (v) Assign each person a random BMI and type 'person'.

- A. $v \rightarrow ii \rightarrow i \rightarrow iv \rightarrow iii$
- B. $ii \rightarrow v \rightarrow i \rightarrow iv \rightarrow iii$
- C. $ii \rightarrow i \rightarrow v \rightarrow iv \rightarrow iii$
- D. $ii \rightarrow v \rightarrow iv \rightarrow i \rightarrow iii$

Answer: B ($ii \rightarrow v \rightarrow i \rightarrow iv \rightarrow iii$)

Reference: Lectures 47–52 – Fatman Model build order

Solution: First create the graph with 100 person nodes (ii), then assign each a random BMI and type (v), then add the five foci (i), then connect each person to one random focus (iv). Only after the city is set up does the simulation loop run — each round applying homophily, closure, then the BMI change (iii).

Q9. [MCQ] To analyse the evolution, the network is saved after each round and reloaded later. Which functions and observation are correct?

- A. Saved with `nx.read_gml`, reloaded with `nx.write_gml`; density tends to fall over the rounds.
- B. Saved only as plain-text edge lists; the network's density stays constant throughout.
- C. Saved with `nx.write_gml`, reloaded with `nx.read_gml`; density tends to rise over the rounds.
- D. Cannot be saved at all, so the network's evolution cannot be analysed afterwards.

Answer: C

Reference: Lecture 53 – Storing and analysing longitudinal data (≈00:41–08:50)

Solution: Each round the graph is written with `nx.write_gml('evolution_t.gml')` and later read back with `nx.read_gml`. Plotting density across the snapshots shows it rising roughly steadily, because homophily and the three closures keep adding edges as the model evolves.

Q10. [MSQ] Which statements about the Fatman model are correct? (Select all that apply.)

- I. The homophily link probability is inversely related to the absolute BMI difference between two people.
 - II. The same formula $1 - (1 - p)^k$ is used for triadic, foci, and membership closure.
 - III. Obesity is measured by counting the person nodes whose name (BMI) equals 40.
 - IV. Social influence is implemented by directly adding friendship edges between any two obese people.
- A. I, II, IV
 - B. II, III, IV
 - C. I, III
 - D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 49–53 – Fatman Model

Solution: Closer BMIs give a higher probability — inverse in the gap (I); one closure formula serves all three closure types (II); and obesity is the count of nodes with BMI (name) equal to 40 (III). Social influence is captured indirectly through gym/eat-out membership changing BMI, not by directly linking obese people, so IV is false.

